

AN INTEGRATED SOUND VELOCITY PROFILING PROBE ARCHITECTURE FOR CLOSED-LOOP ACOUSTIC CALIBRATION, ADAPTIVE SAMPLING, AND SPATIAL CONFIDENCE ESTIMATION

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ABSTRACT

Sound velocity profiling is fundamental to accurate acoustic depth measurement, yet conventional practice treats profile acquisition, system calibration, and quality control as disconnected operations. This paper proposes a methodology for continuous closed-loop validation of sonar depth against mechanical measurement through an integrated probe architecture. The system implements thermal-lag compensation through programmed dwell timing and adaptive sampling triggered by real-time sensor discrepancy detection. A spatial confidence framework is proposed that optimizes the trade-off between survey efficiency and data quality through dynamic interpolation weighting and boundary-calibrated uncertainty estimation. The methodology is hardware-agnostic, optimized for high-gradient freshwater environments, and intended for integration with a wide range of hydrographic survey platforms.

Companion Document Notice:

A companion architectural framework document provides the generalized defensive disclosure and explicitly covers all equivalent methods, sensors, and hardware configurations falling within the scope of the principles described herein.

1. INTRODUCTION

Sound velocity is the primary environmental variable governing the accuracy of acoustic depth measurements in hydrographic surveying. Temperature and pressure gradients along the water column create velocity variations that refract acoustic rays; without accurate correction, these variations introduce systematic depth errors that propagate through the entire bathymetric product.

Standard practice for sound velocity profiling employs dedicated instruments—typically time-of-flight probes that measure sound velocity alongside temperature and pressure. Concurrently, bar-check procedures provide independent mechanical validation of sonar depth calibration, using a physical target lowered to known depths beneath the transducer.

These operations are conventionally performed as separate manual tasks. The SVP probe descends; the bar-check plate descends. Neither informs the other in real-time, and neither adapts its sampling strategy based on the other's measurements.

This paper describes a methodology that closes this loop: a single integrated probe that measures sound velocity while simultaneously providing a physical reference target for the sonar. The system enables real-time cross-validation and adaptive sampling in response to detected gradients. A spatial confidence framework is proposed that qualifies uncertainty across the survey area. This document presents a concrete illustrative implementation; a companion architectural framework document provides the broader legal and methodological foundation covering all equivalent implementations.

2. SCOPE

This paper presents a conceptual architecture for an integrated sound velocity profiling probe and associated survey methodology. The focus is the interaction between sound velocity acquisition, mechanical depth reference, adaptive sampling, and uncertainty management. Specific implementations of mechanical design, electronics, firmware, communication interfaces, and deployment platforms are intentionally left implementation-dependent. The methodology is intended to be adaptable to a wide range of hardware configurations and survey environments.

3. PROBLEM STATEMENT

3.1 The Thermal Lag Artifact

Commercial sound velocity probes achieve high static accuracy—typically ± 0.02 m/s or better. However, these specifications assume the sensor has reached thermal equilibrium with its environment. A thermistor embedded in a metallic housing has thermal mass. When the probe descends through a sharp thermocline—frequent in stratified freshwater bodies—the housing temperature lags behind the surrounding water. The logged sound velocity then corresponds not to the water at the current depth, but to water encountered seconds earlier.

This thermal inertia is rarely addressed in operational protocols. Standard practice is to lower the probe at a fixed rate and log continuously, assuming the readings represent the water at the probe's instantaneous position. The resulting error is systematic and depth-dependent.

3.2 Fixed-Increment Sampling and Missed Gradients

Conventional SVP profiles are collected at fixed depth increments—every meter or every two meters—regardless of the actual gradient structure. If an intense thermocline lies between two sampling points, the profile is linearly interpolated across it, introducing error into every ray-traced beam that passes through that layer. The magnitude of this error scales with the gradient strength and the distance between samples.

3.3 The Manual Bar-Check Bottleneck

The bar-check remains a manual procedure requiring a physical plate lowered on marked cables or chains. It is typically performed once per survey session and treated as a static calibration, yet sound velocity conditions evolve throughout the day. Diurnal heating and solar stratification modify the velocity structure over timescales comparable to a survey operation. A single baseline calibration does not capture this dynamic evolution.

3.4 Spatial Uncertainty

A single SVP profile represents truth only at its exact spatial coordinate and timestamp. The confidence in that profile decays with distance from the measurement point, yet conventional practice treats each cast as globally valid or interpolates linearly without weighting. While geostatistical methods provide frameworks for spatial interpolation, a methodology that integrates a continuously updated confidence field directly into adaptive SVP acquisition and survey planning during the survey itself remains absent from standard practice.

4. PROPOSED ARCHITECTURE: INTEGRATED SOUND VELOCITY PROBE

4.1 Single-Housing Design

The proposed architecture integrates two functions into a single streamlined housing:

- **SVP payload:** A time-of-flight sound velocity sensor with co-located temperature and pressure sensors.
- **Bar-check reference:** A physical target surface that provides a clear acoustic return for the sonar.

This integration eliminates the operational overhead of separate deployments. A single cast provides both the SVP profile and a real-time bar-check at every sampled depth, where the acoustic target surface and the sensor array share a fixed, known geometric relationship.

4.2 Redundant Depth Measurement

The probe is deployed on a tether incorporating a redundant, non-acoustic, non-pressure-dependent displacement or position vector tracking system. This may include a rotary encoder measuring cable payout, physical markers or fiducials on the cable at fixed intervals, or dead-reckoning winch telemetry. This redundant system establishes a physical reference against which the sonar's acoustic depth can be continuously validated.

4.3 Thermal-Lag Compensation

Rather than attempting real-time thermal correction through complex mathematical modeling, the proposed methodology employs a deterministic operational protocol:

- 1 The probe is deployed at a controlled, gentle rate to minimize wake-induced mixing.
- 2 At each programmed sampling depth, the deployment is paused.

- 3 The probe dwells for a duration sufficient for the housing to reach thermal equilibrium with the surrounding water.
- 4 Only after the dwell period is the sound velocity logged.

The dwell duration is selected empirically according to the thermal response characteristics of the sensor package. This approach is hardware-agnostic and does not require real-time modeling of the probe's specific thermal time constant.

4.4 Adaptive Sampling

The system implements a closed-loop sampling strategy:

- 1 The probe profiles at a default increment selected for the water body.
- 2 At each depth, the sonar's measured acoustic depth to the integrated target is compared against the mechanical depth from the reference system.
- 3 If the discrepancy exceeds a configurable tolerance, the system flags that depth interval as a high-gradient zone.
- 4 On a subsequent cast, the probe dynamically alters its sampling parameters within the flagged interval—reducing its sampling increment or adjusting its physical winch velocity—to capture the gradient at higher resolution.

This targets sampling effort precisely where it is needed: at sharp thermoclines and regions of high velocity gradient.

5. DESIGN PHILOSOPHY

- **Hardware Independence:** The methodology is defined by its operational logic, not by specific sensors or platforms. This ensures adaptability across different survey systems.
- **Continuous Validation:** Rather than relying on periodic, manual calibration events, the system continuously cross-validates acoustic measurements against a mechanical reference. This shift from episodic to continuous quality assurance improves data reliability.
- **Explicit Uncertainty:** Uncertainty is treated as a mapped quantity, not as a hidden or post-processed artifact. By making confidence estimates explicit and dynamically updated, the methodology enables informed decision-making during the survey itself.
- **Active Learning:** A subset of planned casts is allocated to boundary validation—empirically checking whether the spatial uncertainty model matches reality. If deviations are detected, the system adapts, concentrating sampling where gradients are strongest.
- **Traceable Audit Trail:** Every measurement is logged with synchronized time, mechanical depth, acoustic depth, and confidence metadata. This provides a complete and defensible record for retrospective quality assessment and certification.

6. SPATIAL CONFIDENCE FRAMEWORK

6.1 *Uncertainty as a Mapped Quantity*

A single SVP cast represents truth only at its exact spatial coordinate and timestamp. For survey areas larger than a single cast point, interpolation is required, but interpolation confidence decays with distance from the measured point.

The proposed framework defines a spatial confidence field for each survey area. At the cast coordinate, confidence equals 1.0. Confidence decays with distance from the cast, modeled using geostatistical, stochastic, or deterministic spatial interpolation techniques—including exponential or Gaussian decay functions optimized in real-time by boundary constraints. At the boundary between adjacent cast cells, confidence is the interpolated value between their respective confidence fields.

6.2 *Boundary Casts as Spatial Covariance Sampling*

To calibrate the confidence decay function for each specific water body, the framework allocates a subset of planned casts to boundary regions between grid cells. These boundary casts directly measure the horizontal SVP gradient, providing empirical data on how much the profile changes over distance. The measured boundary gradient dynamically adjusts the correlation length of the confidence decay function for that specific survey area, enabling real-time adjustments to the survey plan.

6.3 *Temporal Decoupling*

Survey geometry may incorporate interlaced passes at different headings to provide spatial redundancy and temporal decoupling. Passes conducted at different times capture the diurnal thermal evolution of the water column. Each pass includes its own SVP casts, and the final bathymetric product is stitched using temporally appropriate profiles.

7. OPERATIONAL PROTOCOL

7.1 *Cast Sequence*

- 1 The platform positions at the cast coordinate and holds station.
- 2 The probe is deployed at a controlled rate.
- 3 At each programmed depth increment: the deployment is paused; the probe dwells for thermal equalization; the SVP logs sound velocity, temperature, and pressure; the sonar logs the acoustic depth to the probe's bar-check target; the mechanical depth measurement system logs reference depth; the system compares acoustic versus mechanical depth and flags the interval if the discrepancy exceeds tolerance.
- 4 The probe is retrieved. The continuous data collected during the ascent phase may be cross-referenced against the descent profile as a high-speed anomaly scan to detect rapid temporal changes (e.g., internal waves or moving thermal fronts) before discarding.

7.2 Closed-Loop Response

If a cast identifies a high-gradient interval:

- The system schedules a supplemental high-resolution cast at the same coordinate.
- The supplemental cast samples the flagged depth band at reduced increments.
- The high-resolution data replaces interpolated values for that interval.
- The trigger threshold is configurable based on the required survey order precision.

8. SYSTEM INTEGRATION

8.1 Platform Compatibility

The methodology is intended for integration with a variety of shallow-water and freshwater survey platforms, including autonomous surface vessels, towed systems, and vessel-mounted deployments. Its core value lies in the operational methodology—closed-loop validation between mechanical and acoustic depth—rather than in any specific hardware implementation.

8.2 Data Architecture

The system maintains a timestamped, append-only data ledger design for real-time edge processing. Telemetry packets are structured to decouple raw environmental sampling from the adaptive control loop, allowing seamless integration via industry-standard protocols without introducing processing latency to the primary navigation stack. Every measurement is inherently associated with:

- Synchronized time.
- Mechanical depth.
- Acoustic depth.
- Sound velocity, temperature, and pressure data.
- Confidence metadata.

9. DISCUSSION

9.1 Operational Efficiency

Conventional practice separates SVP profiling and bar-checking into distinct manual operations. The proposed methodology reduces station time by eliminating separate bar-check deployments, automating the calibration validation loop, and targeting sampling effort only where aggressive thermal gradients require it. For autonomous surface vessels (ASVs) operating under strict energy constraints, eliminating standalone calibration events directly optimizes battery life and maximizes survey line-mile production per deployment window.

9.2 Data Quality and Auditability

The mechanical-acoustic cross-validation provides a continuous audit trail. Every depth measurement has an associated mechanical reference, discrepancy flags are securely logged, and the confidence framework makes uncertainty explicit. This shifts the quality assurance burden from post-processed statistical estimation to deterministic, real-time verification capable of supporting strict engineering-grade data requirements.

10. CONCLUSION

This paper has presented a conceptual architecture intended to establish a framework for future validation and refinement of autonomous hydrographic calibration. The methodology addresses persistent limitations in conventional freshwater practice by combining SVP measurement with a physical bar-check reference in a single housing, implementing thermal-lag compensation through programmed dwell timing, and enabling adaptive sampling triggered by real-time mechanical-acoustic discrepancy detection.

The spatial confidence framework treats uncertainty as a mapped quantity, providing a systematic approach to interpolation weighting and survey planning. Experimental verification through controlled freshwater field deployments remains the immediate direction for future implementation.

11. REFERENCES

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INTELLECTUAL PROPERTY NOTICE / PRIOR ART DISCLOSURE

This document constitutes a public, defensive disclosure of a specific, illustrative implementation of a conceptual methodology and system architecture. The functional logic, adaptive sampling triggers, closed-loop calibration loops, and uncertainty frameworks described herein are intended to remain in the public domain as prior art to prevent the issuance of restrictive patents on these methods.

A companion framework document—providing the generalized architectural disclosure and explicitly covers all equivalent methods, sensors, algorithms, and hardware configurations—is published concurrently under the same title (see companion repository document). Together, these documents establish comprehensive prior art against restrictive patents on any system embodying the closed-loop, adaptive, or confidence-mapping principles described herein.